Major project

Objective:

The goal of this project is to create a chatbot using Amazon Lex that helps users book hotel rooms. The chatbot will ask for details like location, room type, check-in date, and number of days. It will also give updates while processing the booking and let the user know if it takes too long.

1)All the information must be conveyed to the user after booking the room and

must be informed to the user the price of the hotel room and the day of stay.

2)Using this chatbot user must be aware of the types of Available rooms

(Classic, Duplex etc)-Choose your own Category as well.

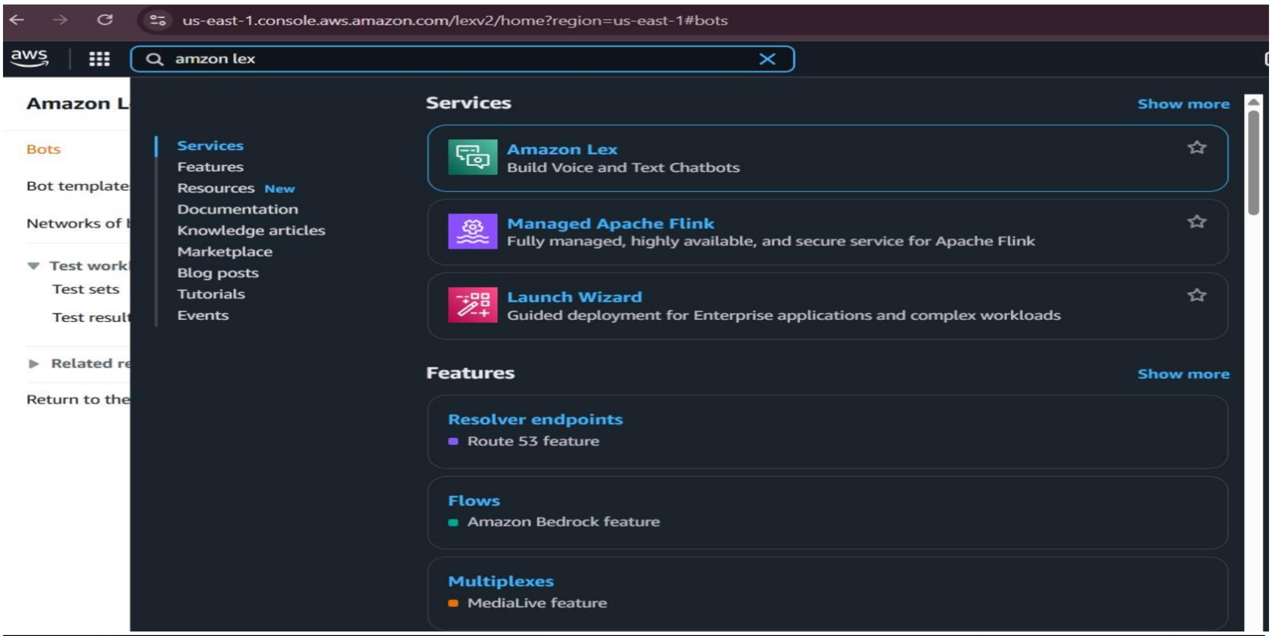
3)All events must be in flow for the fulfillment of the intent.

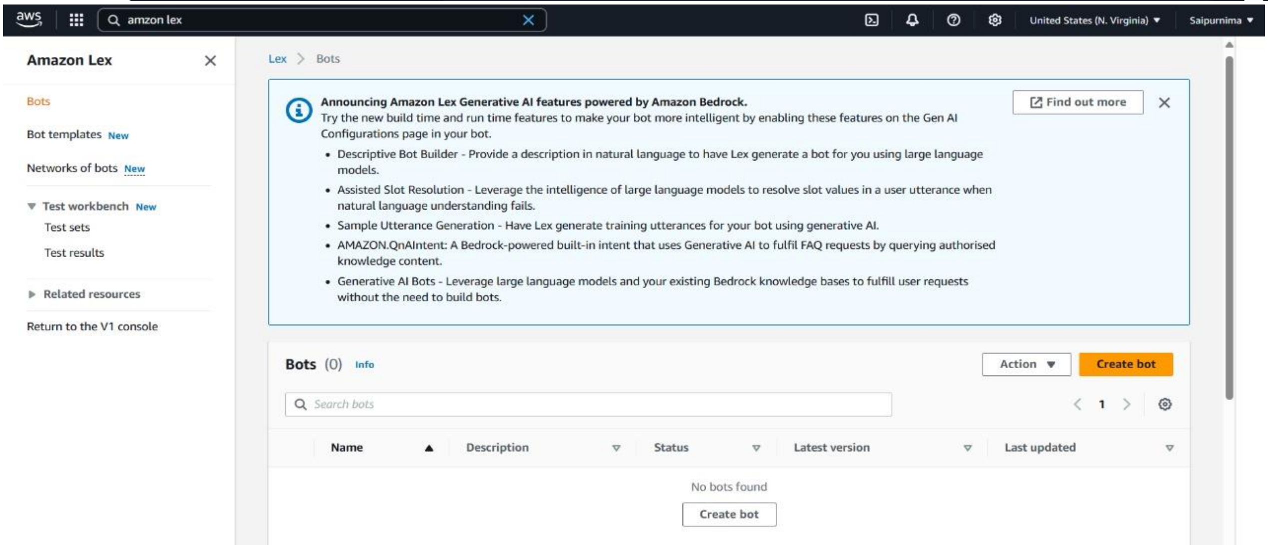
Creation:

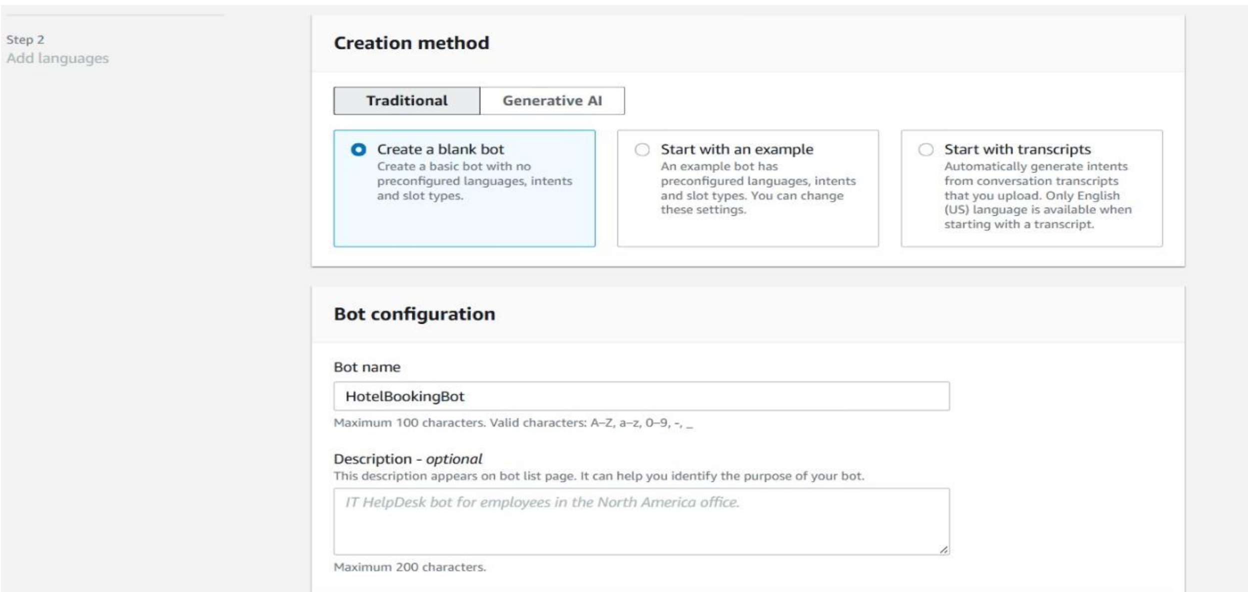
First, log in to your AWS account and open Amazon Lex at https://console.aws.amazon.com/lexv2. Click "Create bot."

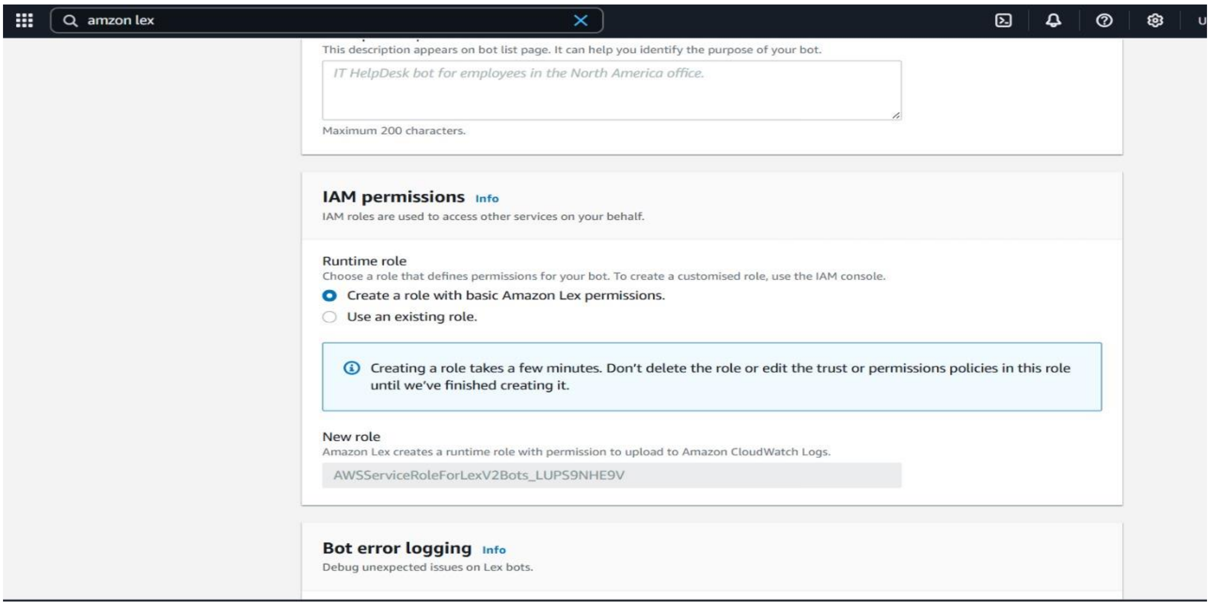
Give your bot the name HotelBookingBot.
When asked about IAM permissions, allow Lex to create and use a new role.
For the COPPA question, select No.

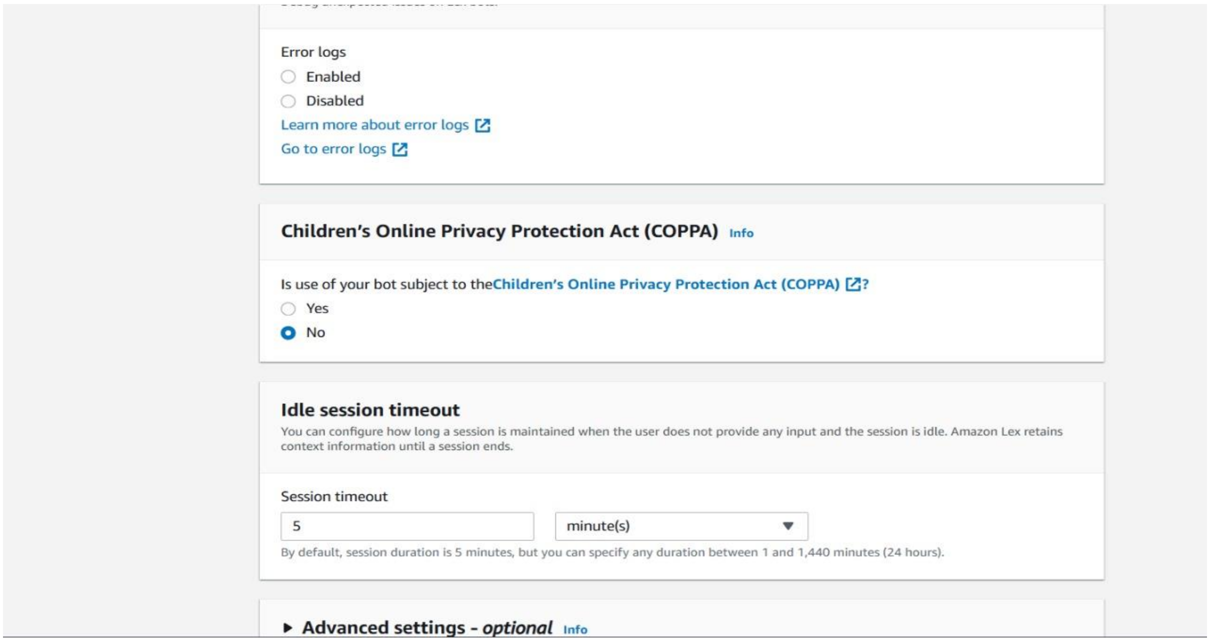
Click Next, then choose English (US) as the language.
Finally, click Done to finish creating the bot.

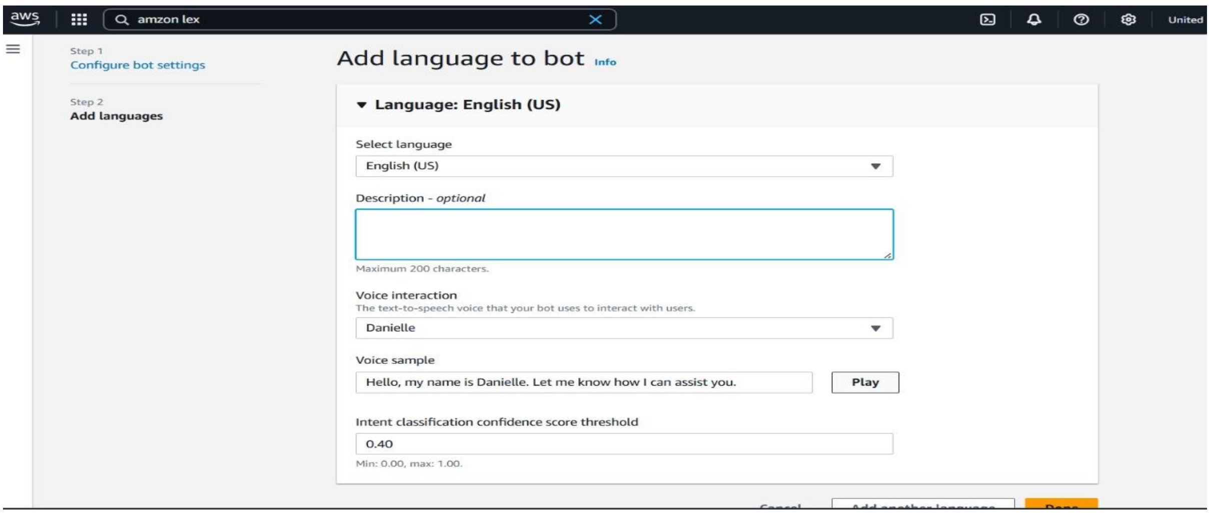

us-east-1.console.aws.amazon.com/lexv2/home?region=us-east-1#bots
aws
amzon lex
Amazon L
Bots
Bot template
Networks of
Test work
Test sets
Test result
Related re
Return to the
Services
Features
Resources New
Documentation
Knowledge articles
Marketplace
Blog posts
Tutorials
Events
Services
Show more
Amazon Lex
Build Voice and Text Chatbots
Managed Apache Flink
Fully managed, highly available, and secure service for Apache Flink
Launch Wizard
Guided deployment for Enterprise applications and complex workloads
Features
Show more
Resolver endpoints
Route 53 feature
Flows
Amazon Bedrock feature
Multiplexes
MediaLive feature


aws
amzon lex
United States (N. Virginia)
Saipurnima
Amazon Lex
Bots
Bot templates New
Networks of bots New
Test workbench New
Test sets
Test results
Related resources
Return to the V1 console
Lex > Bots
Announcing Amazon Lex Generative AI features powered by Amazon Bedrock.
Try the new build time and run time features to make your bot more intelligent by enabling these features on the Gen AI Configurations page in your bot.
Descriptive Bot Builder - Provide a description in natural language to have Lex generate a bot for you using large language models.
Assisted Slot Resolution - Leverage the intelligence of large language models to resolve slot values in a user utterance when natural language understanding fails.
Sample Utterance Generation - Have Lex generate training utterances for your bot using generative AI.
AMAZON.QnAIntent: A Bedrock-powered built-in intent that uses Generative AI to fulfil FAQ requests by querying authorised knowledge content.
Generative AI Bots - Leverage large language models and your existing Bedrock knowledge bases to fulfill user requests without the need to build bots.
Find out more
Bots (0) Info
Action
Create bot
Search bots
1
Name
Description
Status
Latest version
Last updated
No bots found
Create bot

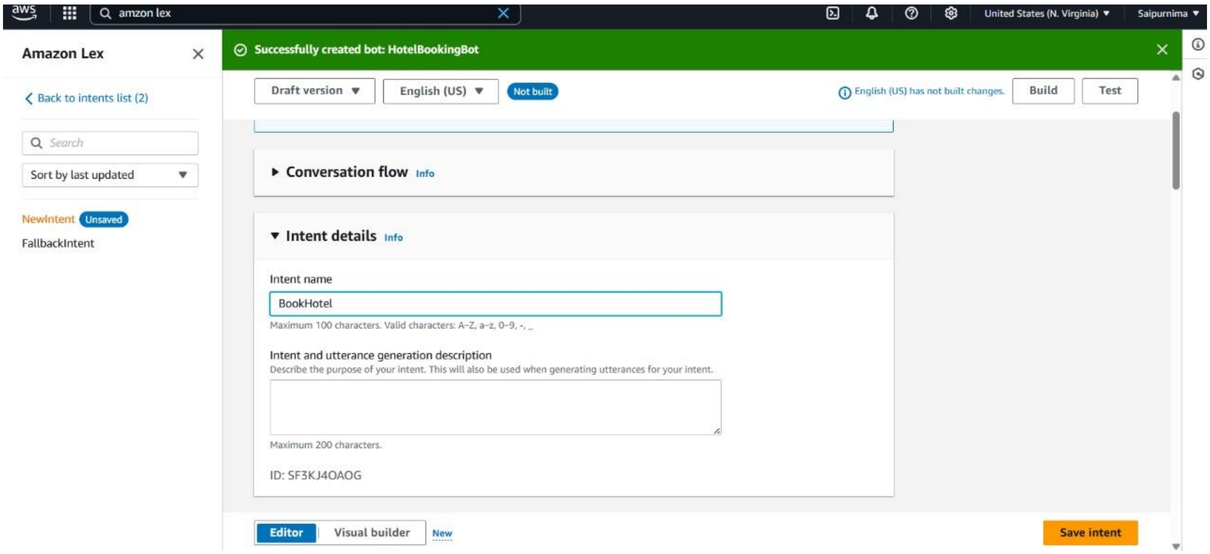

Step 2
Add languages
Creation method
Traditional
Generative AI
Create a blank bot
Create a basic bot with no preconfigured languages, intents and slot types.
Start with an example
An example bot has preconfigured languages, intents and slot types. You can change these settings.
Start with transcripts
Automatically generate intents from conversation transcripts that you upload. Only English (US) language is available when starting with a transcript.
Bot configuration
Bot name
HotelBookingBot
Maximum 100 characters. Valid characters: A-Z, a-z, 0-9, -, _
Description - optional
This description appears on bot list page. It can help you identify the purpose of your bot.
IT HelpDesk bot for employees in the North America office.
Maximum 200 characters.


amzon lex
This description appears on bot list page. It can help you identify the purpose of your bot.
IT HelpDesk bot for employees in the North America office.
Maximum 200 characters.
IAM permissions Info
IAM roles are used to access other services on your behalf.
Runtime role
Choose a role that defines permissions for your bot. To create a customised role, use the IAM console.
Create a role with basic Amazon Lex permissions.
Use an existing role.
Creating a role takes a few minutes. Don't delete the role or edit the trust or permissions policies in this role until we've finished creating it.
New role
Amazon Lex creates a runtime role with permission to upload to Amazon CloudWatch Logs.
AWSServiceRoleForLexV2Bots_LUPS9NHE9V
Bot error logging Info
Debug unexpected issues on Lex bots.


Error logs
Enabled
Disabled
Learn more about error logs
Go to error logs
Children's Online Privacy Protection Act (COPPA) Info
Is use of your bot subject to theChildren's Online Privacy Protection Act (COPPA) ?
Yes
No
Idle session timeout
You can configure how long a session is maintained when the user does not provide any input and the session is idle. Amazon Lex retains context information until a session ends.
Session timeout
5
minute(s)
By default, session duration is 5 minutes, but you can specify any duration between 1 and 1,440 minutes (24 hours).
Advanced settings - optional Info


aws
amzon lex
United
Step 1
Configure bot settings
Step 2
Add languages
Add language to bot Info
Language: English (US)
Select language
English (US)
Description - optional
Maximum 200 characters.
Voice interaction
The text-to-speech voice that your bot uses to interact with users.
Danielle
Voice sample
Hello, my name is Danielle. Let me know how I can assist you.
Play
Intent classification confidence score threshold
0.40
Min: 0.00, max: 1.00.

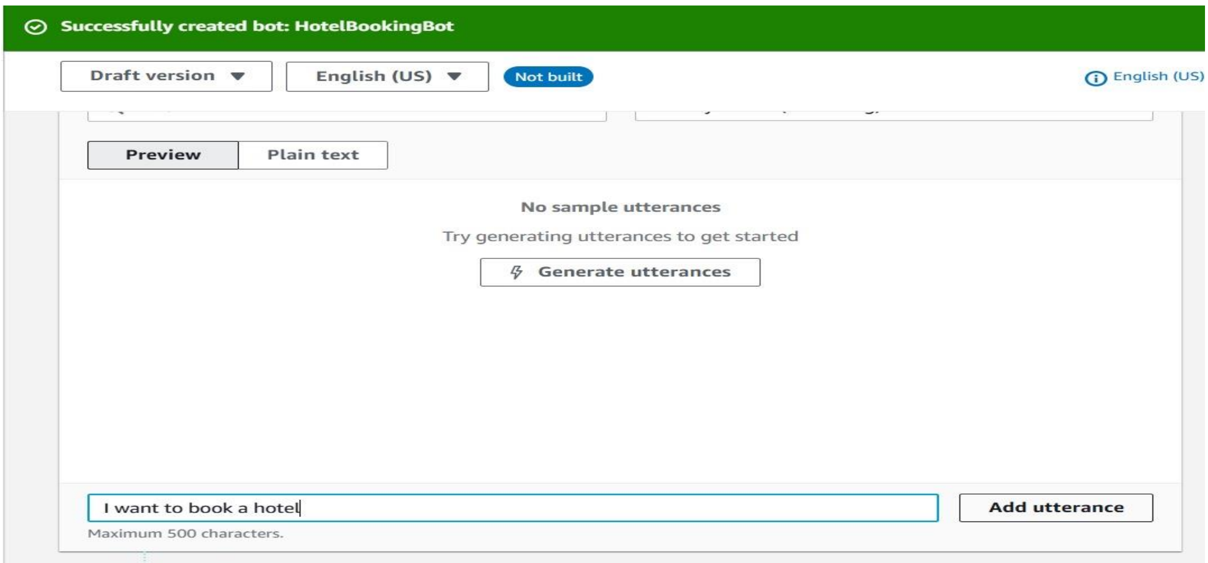
In the **Intent name** field (as shown in your screenshot), replace NewIntent with BookHotel.

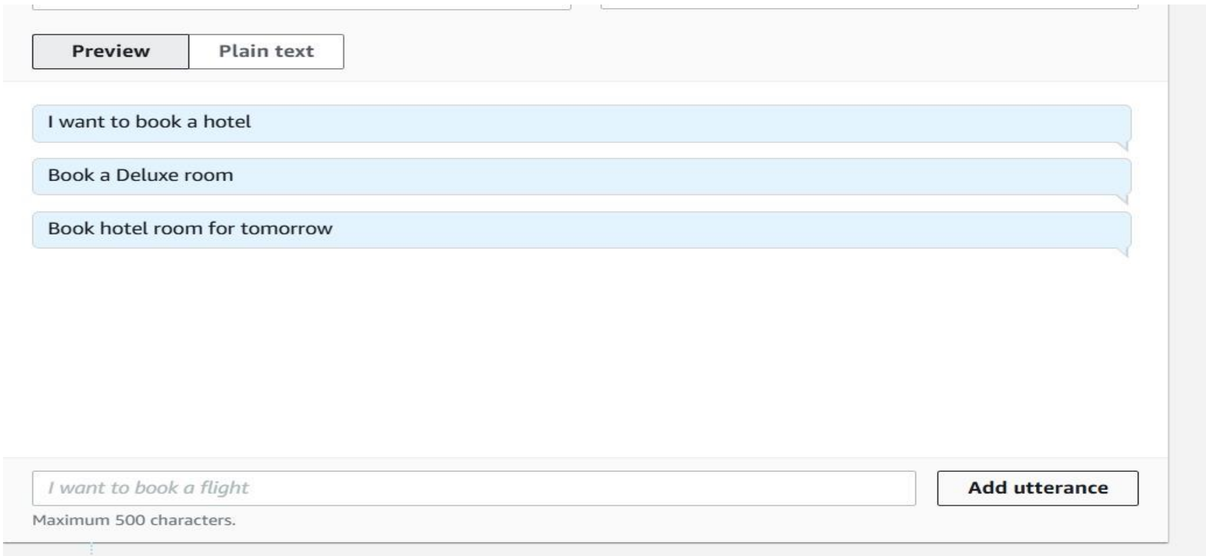


Scroll down to **Sample utterances** (or click "Conversation flow" if collapsed):

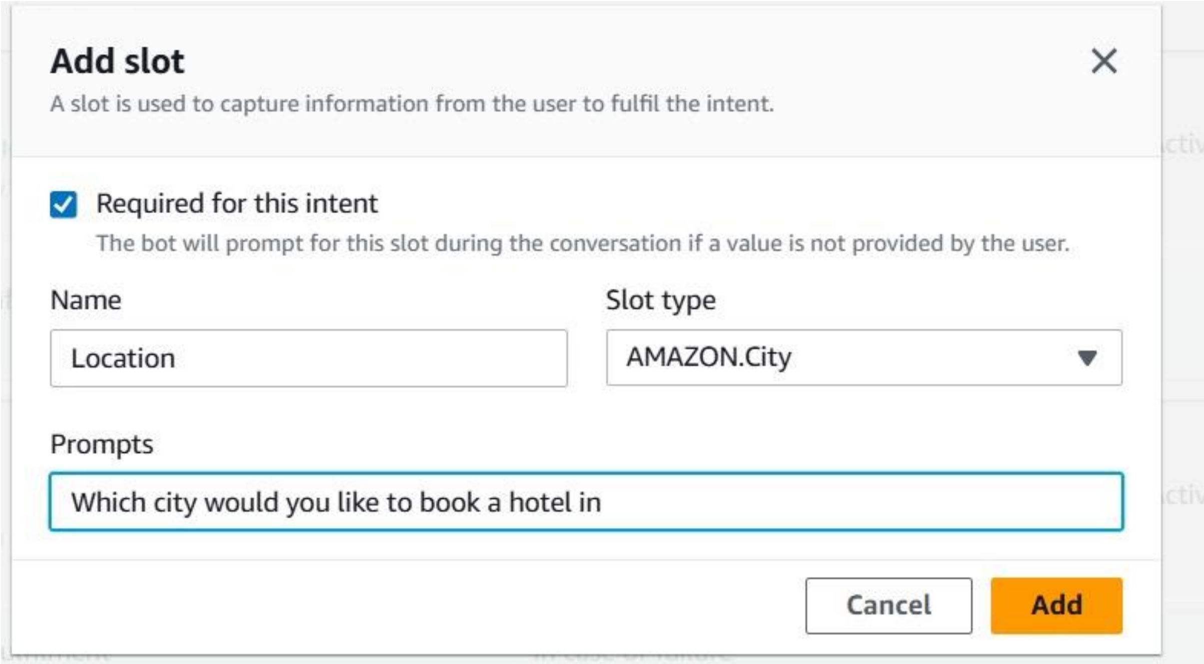
Add examples like:

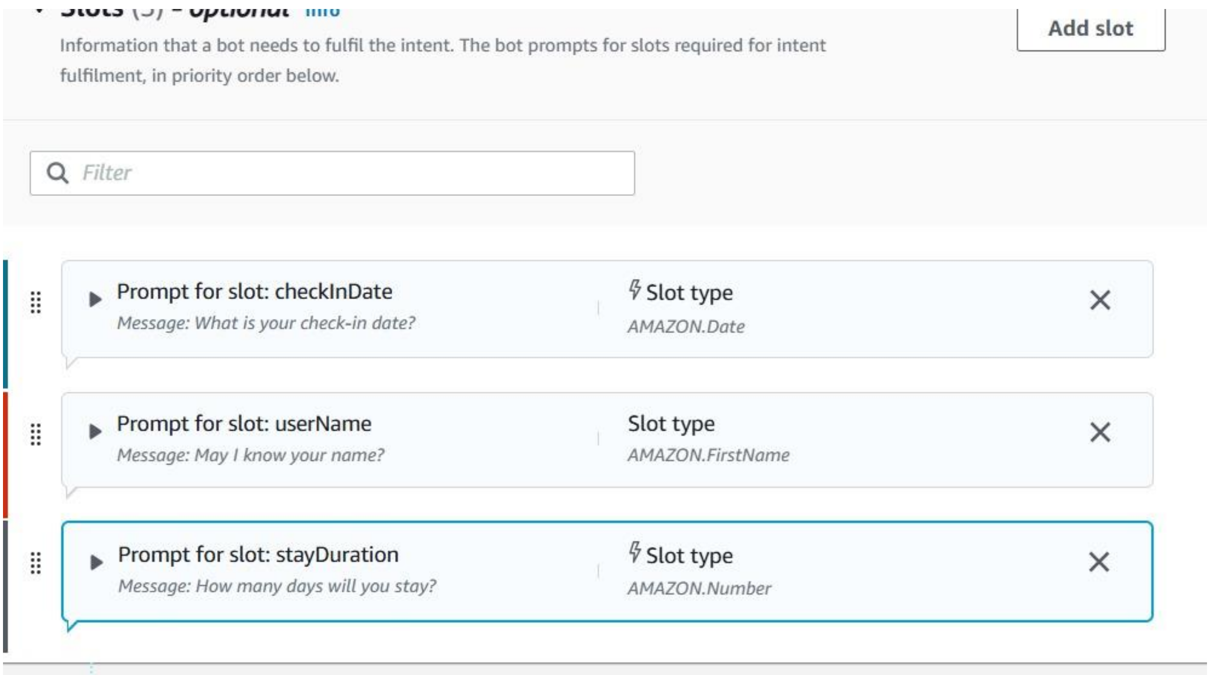
- I want to book a hotel

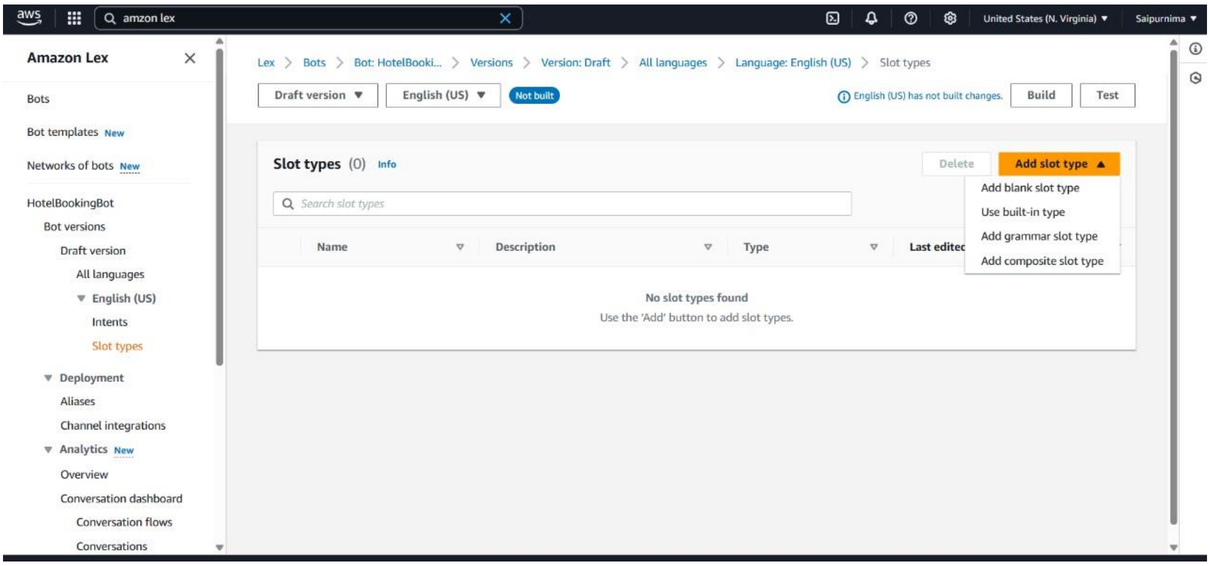




Scroll to the **Slot elicitation** section and add the following slots:



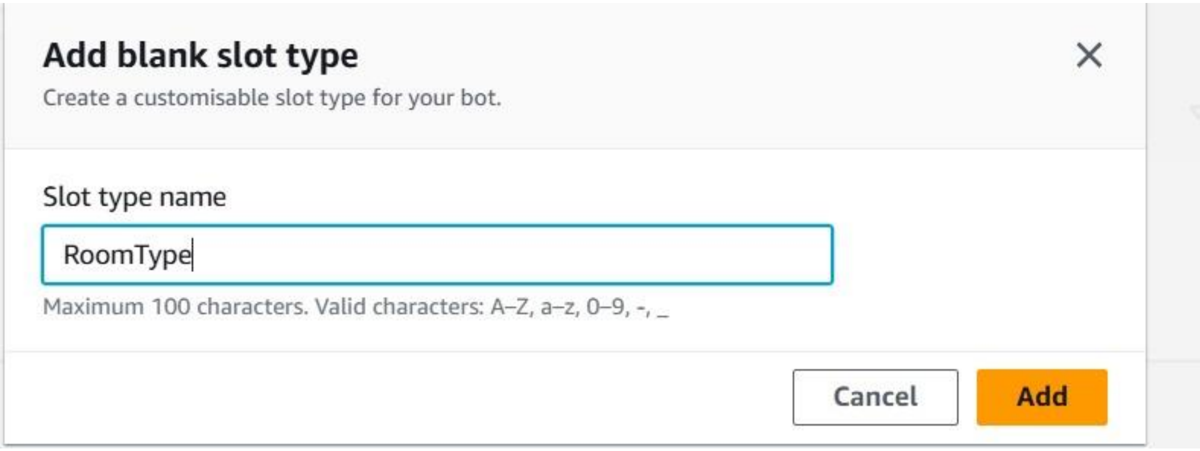


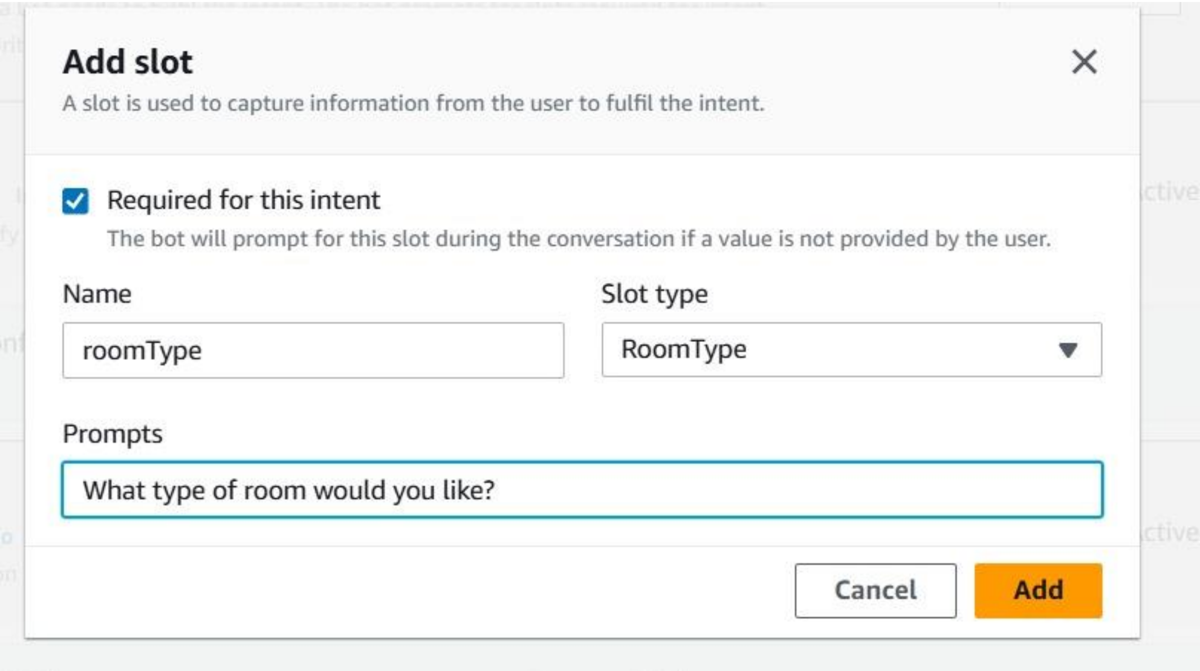


Step 3: Add it to Your Intent

Now go back to Intents > New Intent → Add a new slot:

- **Slot name**: roomType
- **Slot type**: Select RoomType (the one you just created)
- **Prompt**: What type of room would you like?





On the **left menu**, click **Slot types** (under your language — like "English (US)")

Click the **"+ Add slot type"** button

In the **Name** field, type:

Now add these **room types one by one**:

Classic

Deluxe

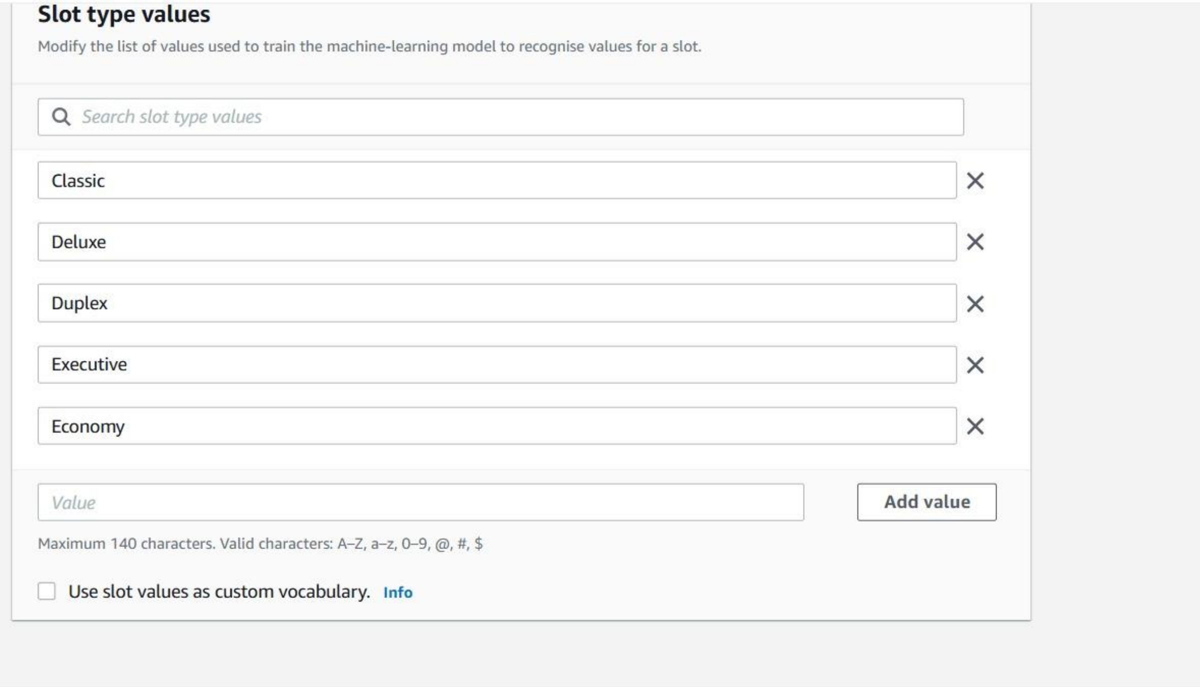
Suite

Duplex

Economy

Executive

After adding all 6 room types, click **Save** (top right)



Step 1: Scroll Down and Click Create Function

Once you've entered:

- Function name: HotelBookingLambda
- Runtime: Node.js 22.x (or Python 3.12 if you prefer Python)

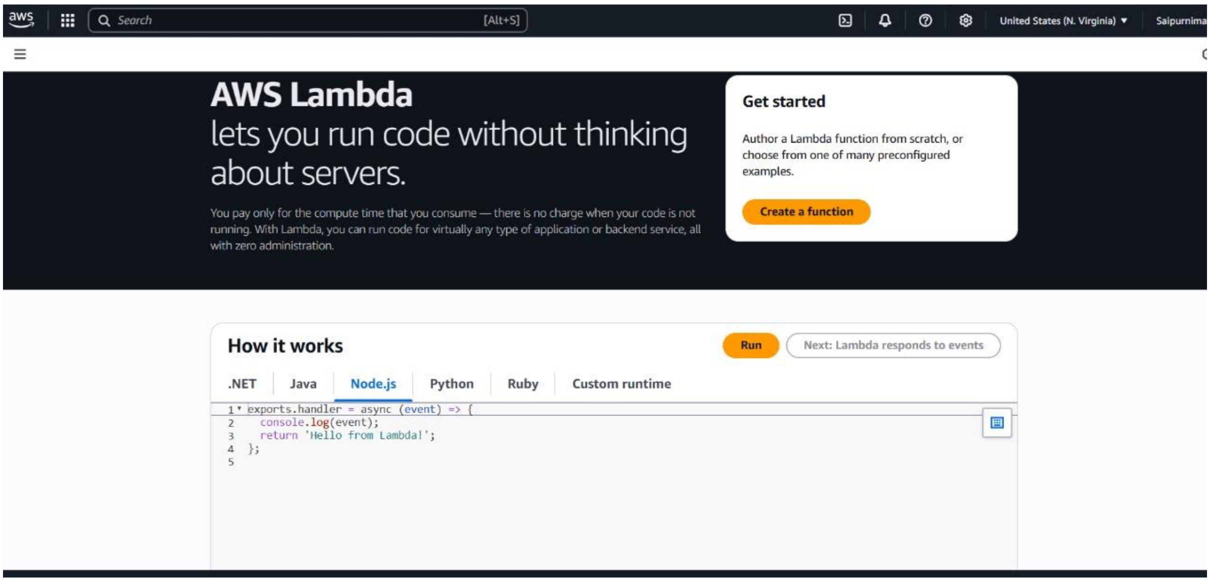
Scroll down and click the **Create function** button at the bot

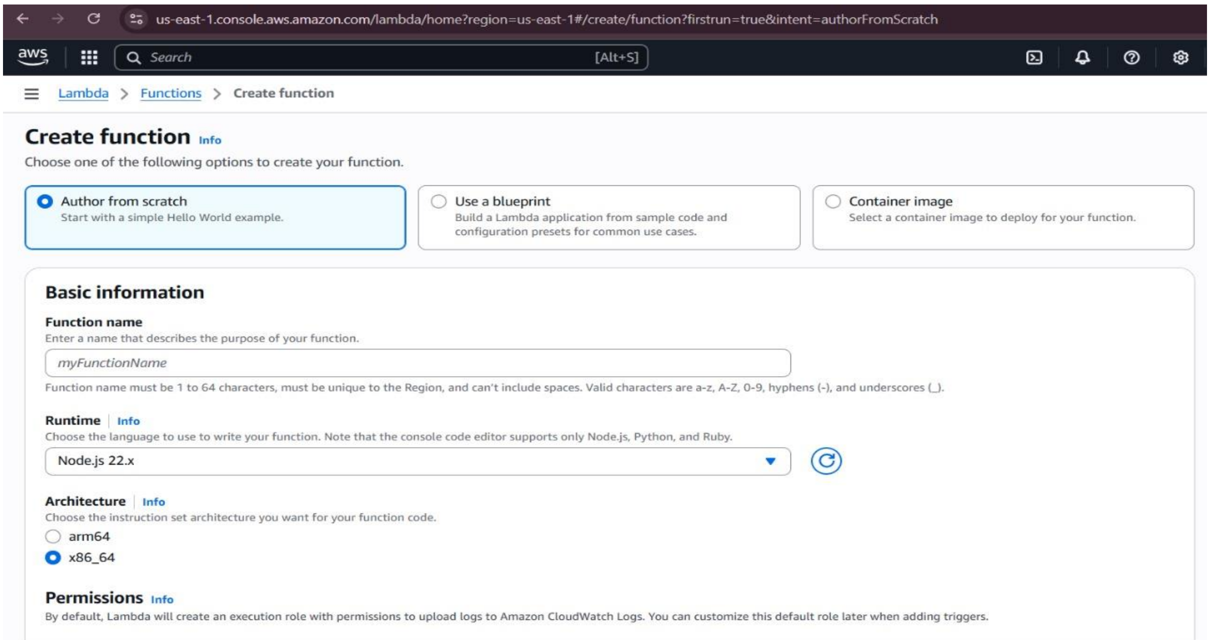
Step 2: Add Lambda Code

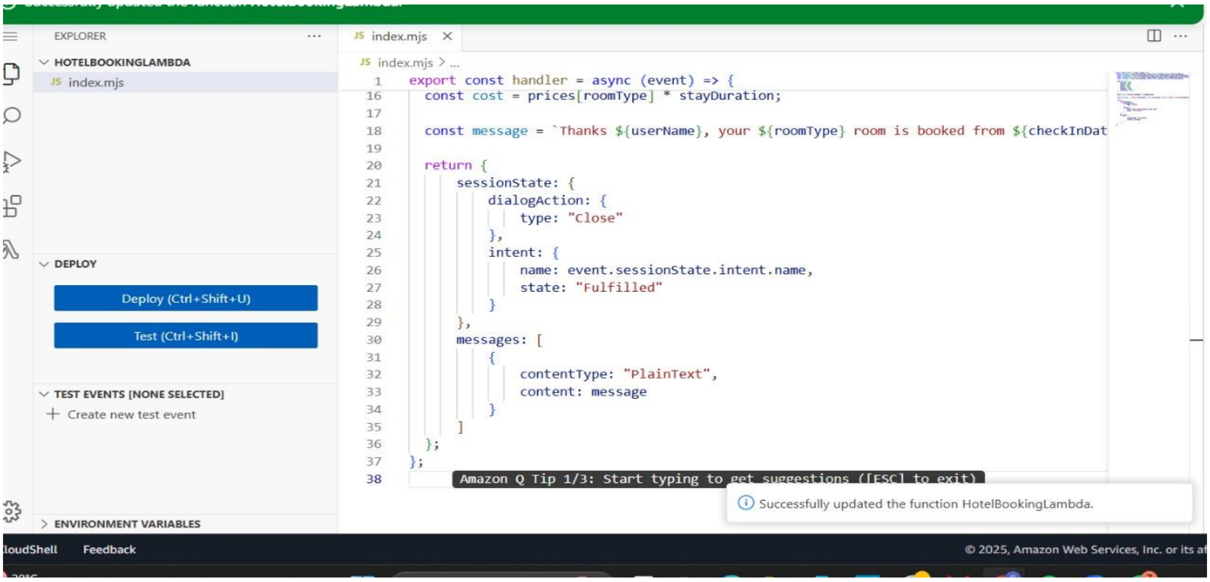
After it's created, you'll land on the function's configuration page. Now do this:

1. Scroll down to the **Code source** section
2. Click **Edit** (pencil icon or directly in editor)

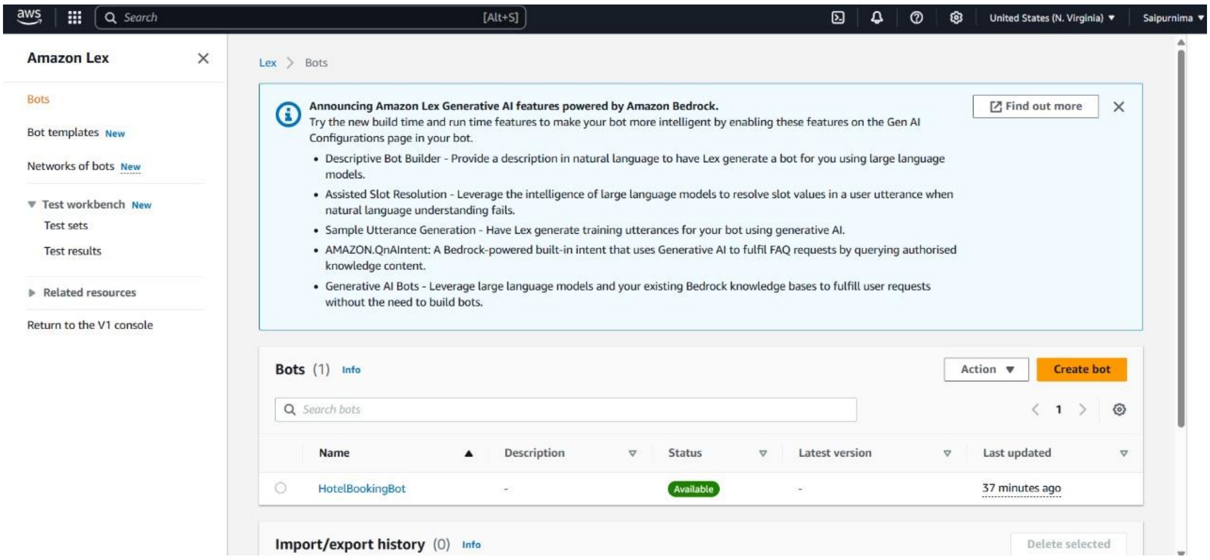
3. Replace the default code with this sample (Node.js version):





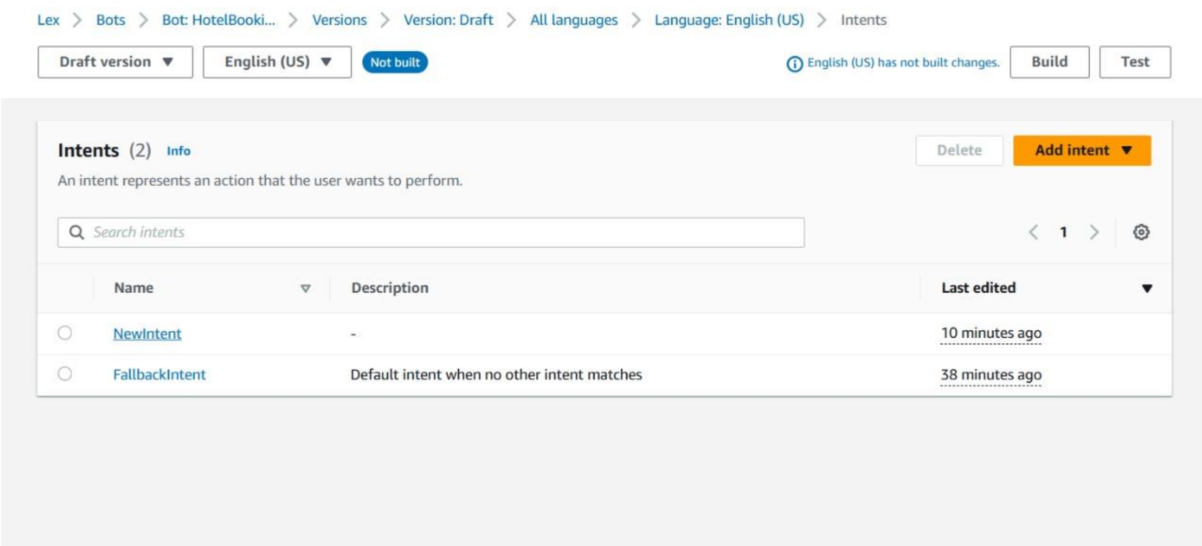


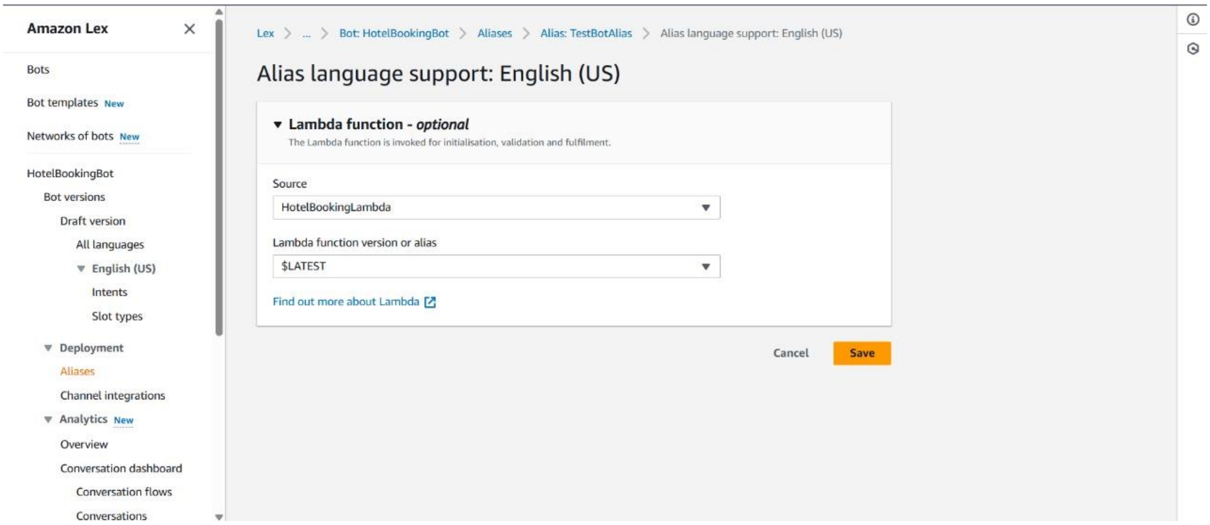
Click the **Deploy** button (top right of code editor) to save and activate your Lambda function.

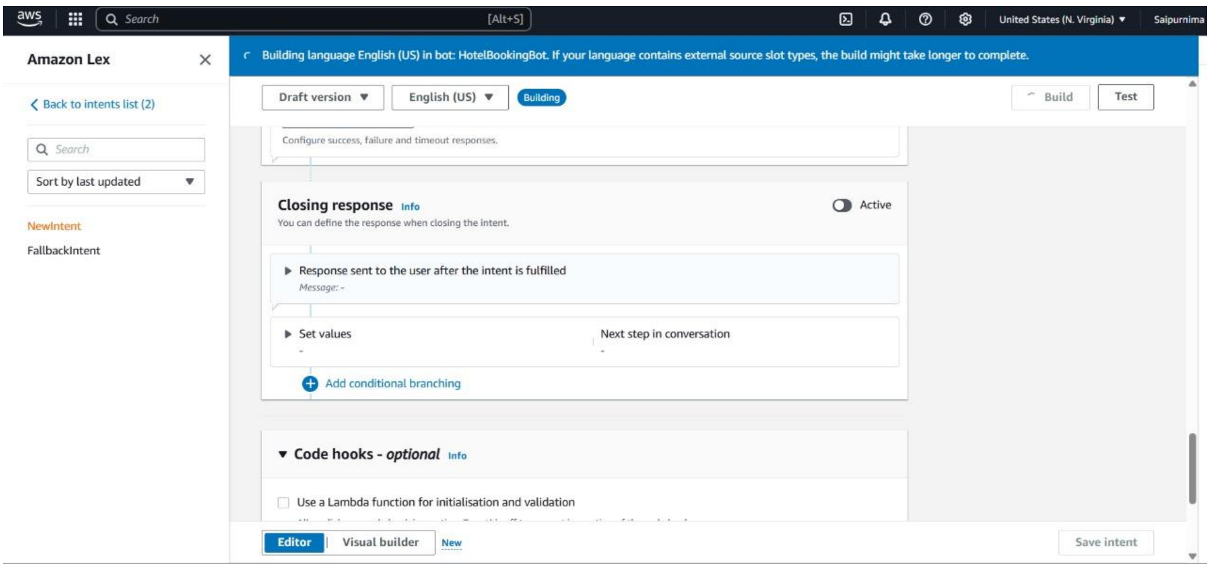


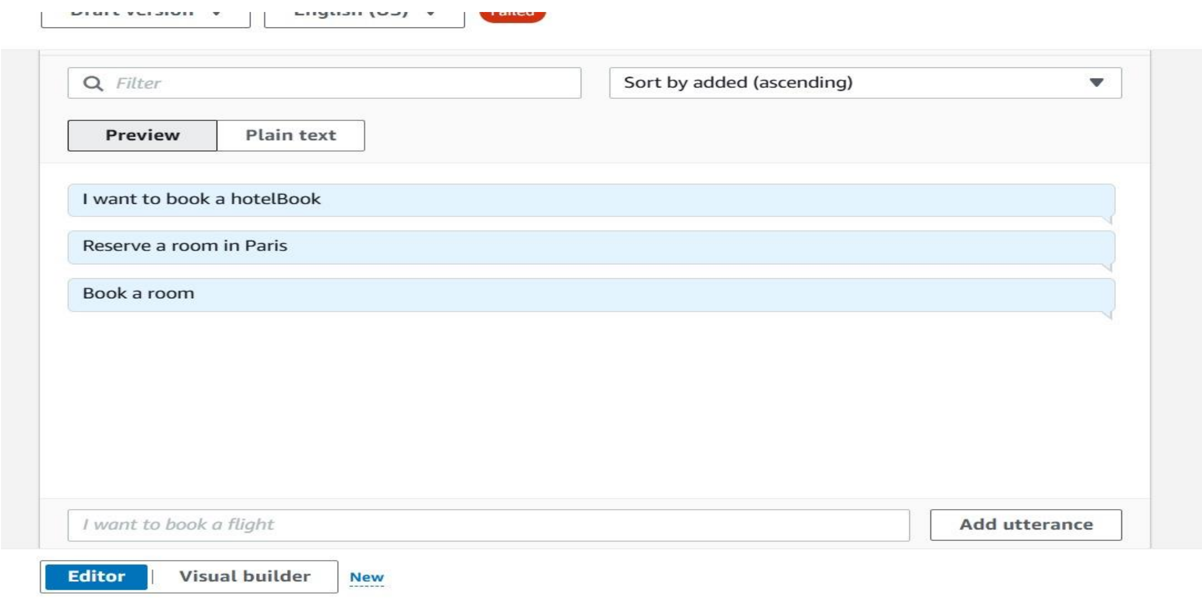
Now, go back to **Amazon Lex**:

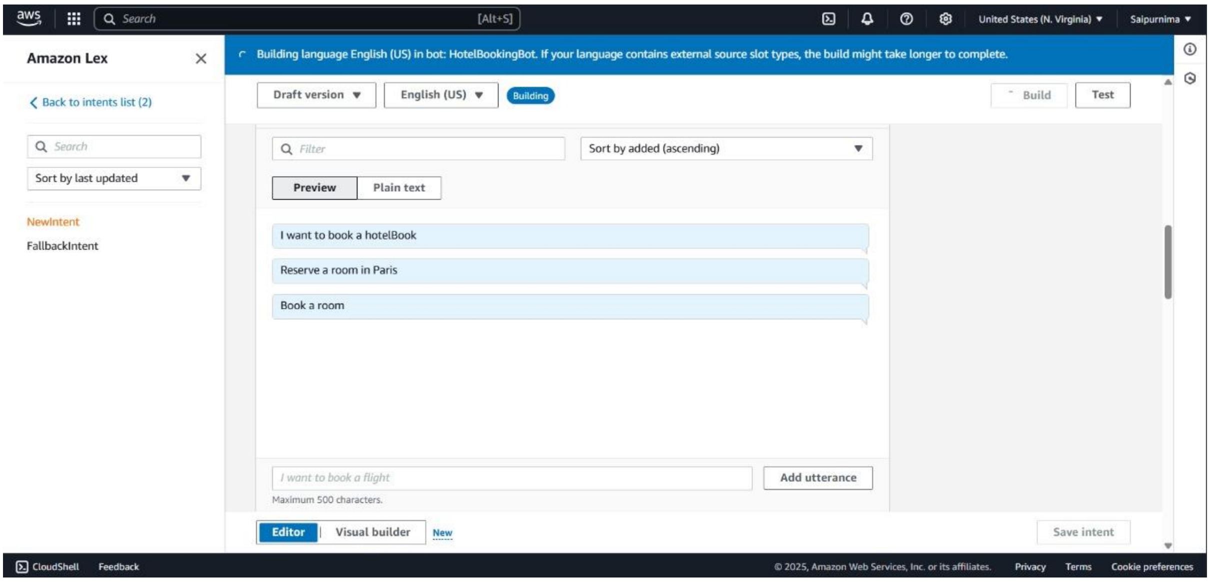
1. Open your Book Hotel intent
2. Scroll to the bottom where it says **Fulfillment**
3. Choose **AWS Lambda function**
4. Select Hotel Booking Lambda
5. Click **Save intent**
6. Then click **Build bot**

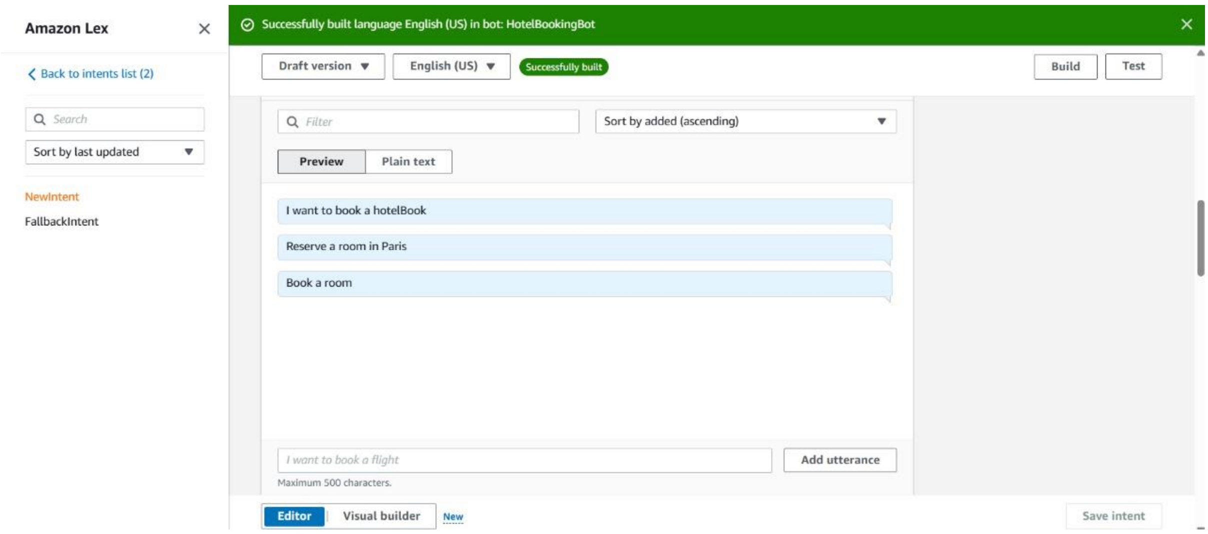

Lex > Bots > Bot: HotelBooki... > Versions > Version: Draft > All languages > Language: English (US) > Intents
Draft version
English (US)
Not built
English (US) has not built changes.
Build
Test
Intents (2) Info
Delete
Add intent
An intent represents an action that the user wants to perform.
Search intents
1
Name
Description
Last edited
NewIntent
-
10 minutes ago
FallbackIntent
Default intent when no other intent matches
38 minutes ago

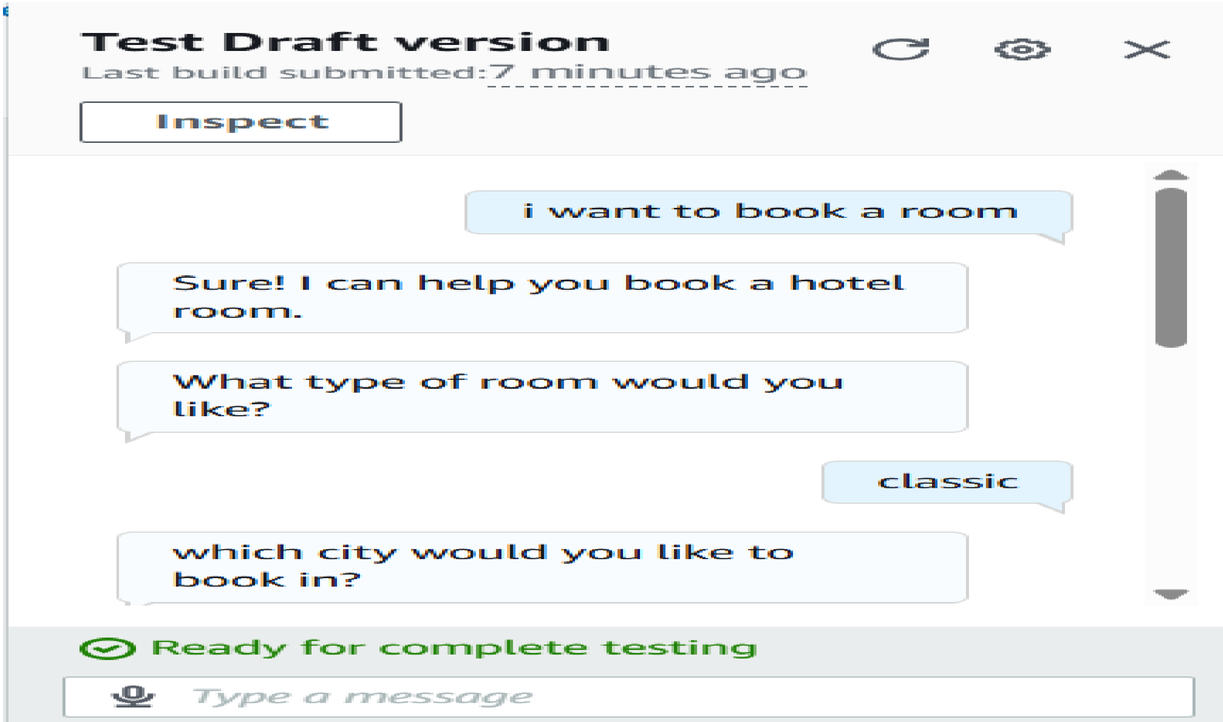

Amazon Lex
Bots
Bot templates New
Networks of bots New
HotelBookingBot
Bot versions
Draft version
All languages
English (US)
Intents
Slot types
Deployment
Aliases
Channel integrations
Analytics New
Overview
Conversation dashboard
Conversation flows
Conversations
Lex > ... > Bot: HotelBookingBot > Aliases > Alias: TestBotAlias > Alias language support: English (US)
Alias language support: English (US)
Lambda function - optional
The Lambda function is invoked for initialisation, validation and fulfilment.
Source
HotelBookingLambda
Lambda function version or alias
$LATEST
Find out more about Lambda
Cancel
Save

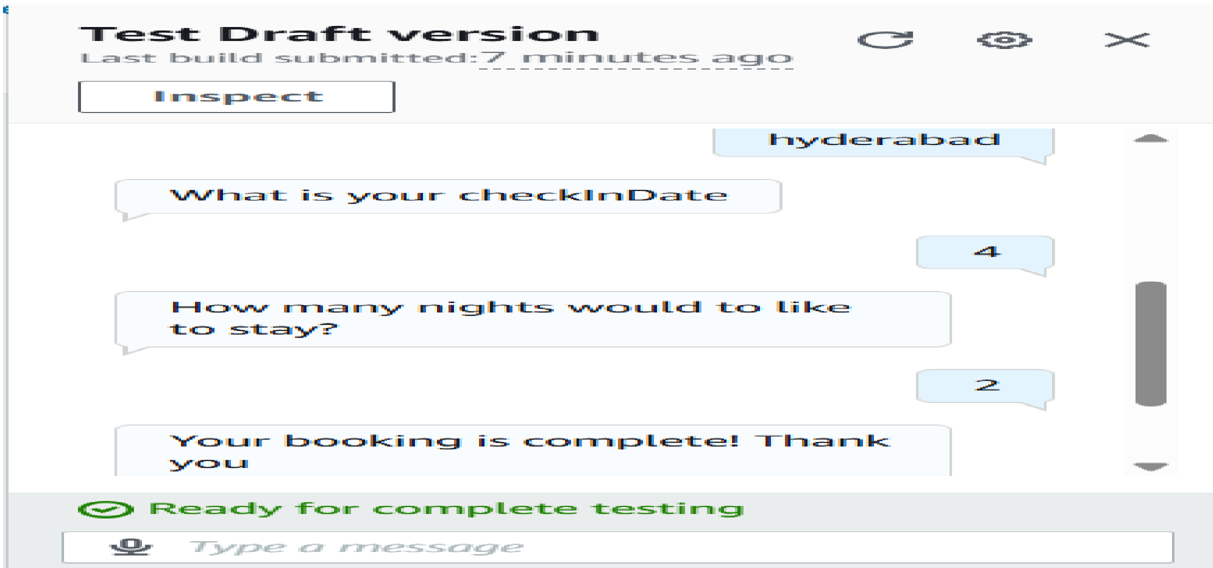

Search
[Alt+S]
United States (N. Virginia)
Saipurnima
Amazon Lex
Building language English (US) in bot: HotelBookingBot. If your language contains external source slot types, the build might take longer to complete.
Back to intents list (2)
Search
Sort by last updated
NewIntent
FallbackIntent
Draft version
English (US)
Building
Build
Test
Configure success, failure and timeout responses.
Closing response Info
Active
You can define the response when closing the intent.
Response sent to the user after the intent is fulfilled
Message: -
Set values
Next step in conversation
Add conditional branching
Code hooks - optional Info
Use a Lambda function for initialisation and validation
Editor
Visual builder
New
Save intent


Filter
Sort by added (ascending)
Preview
Plain text
I want to book a hotelBook
Reserve a room in Paris
Book a room
I want to book a flight
Add utterance
Editor
Visual builder
New


Search
[Alt+S]
United States (N. Virginia)
Saipurnima
Amazon Lex
Building language English (US) in bot: HotelBookingBot. If your language contains external source slot types, the build might take longer to complete.
Back to intents list (2)
Draft version
English (US)
Building
Build
Test
Search
Sort by last updated
NewIntent
FallbackIntent
Filter
Sort by added (ascending)
Preview
Plain text
I want to book a hotelBook
Reserve a room in Paris
Book a room
I want to book a flight
Add utterance
Maximum 500 characters.
Editor
Visual builder
New
Save intent
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Amazon Lex
Successfully built language English (US) in bot: HotelBookingBot
Back to intents list (2)
Draft version
English (US)
Successfully built
Build
Test
Search
Sort by last updated
NewIntent
FallbackIntent
Filter
Sort by added (ascending)
Preview
Plain text
I want to book a hotelBook
Reserve a room in Paris
Book a room
I want to book a flight
Add utterance
Maximum 500 characters.
Editor
Visual builder
New
Save intent


Test Draft version
Last build submitted:7 minutes ago
Inspect
i want to book a room
Sure! I can help you book a hotel room.
What type of room would you like?
classic
which city would you like to book in?
Ready for complete testing
Type a message


Test Draft version
Last build submitted:7 minutes ago
Inspect
hyderabad
What is your checkInDate
4
How many nights would to like to stay?
2
Your booking is complete! Thank you
Ready for complete testing
Type a message